Shirpur Education Society's

R. C. PATEL INSTITUTE OF TECHNOLOGY, SHIRPUR

An Autonomous Institute

(Affiliated to Dr. Babasaheb Ambedkar Technological University, Lonere)

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(स्वायत्त महाविद्यालय)

Programme: B. Tech (AI & DS)/B. Tech (AI & ML)

Year: III/Semester V (Exam Year: 2025-2026)

Subject: Machine Learning **Time:** 09:30 am - 11:30 am (02:00 Hrs.)

Date: 03 Dec 2025 **Max Marks:** 60

END SEMESTER EXAMINATION ODD SEM- V -2025-REGULAR

Instructions:
1. This question paper contains **3** pages
2. Answer to each new question to be started on a fresh page.
3. Figure in right hand side indicates full marks
4. Assume suitable data wherever required, but justify it.
5. Support your answers with neat labelled diagrams, wherever necessary

1.A. Explain the concept of Bias-Variance Trade-off with an example illustrating overfitting and underfitting. **5**

1.B. Analyze how inductive bias influences the learning process and impacts the generalization ability of a machine learning model. **5**

1.C. (Attempt Any 1 Question) **5**

i. Explain the concept of the margin in SVM. **5**

ii. Differentiate between **Linear, Polynomial**, and **Radial Basis Function (RBF)** kernels used in SVM. Provide examples of situations where each kernel is most suitable. **5**

2.A. **Illustrate the concepts of overfitting and underfitting. How can the issue of overfitting be resolved?** **5**

2.B. (Attempt Any 1 Question) **10**

i. Obtain the equation of best fit line for the given dataset. Predict the glucose level for the person having age 55 years. **10**

AGE	43	21	25	42	57	59
GLUCOSE LEVEL	99	65	79	75	87	81

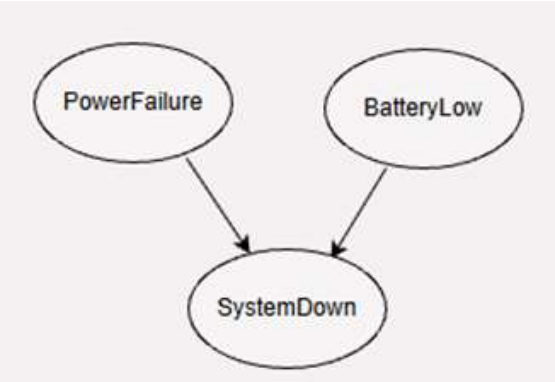
ii. Apply GINI index method to determine the root node for building the decision tree for the given dataset. **10**

Weather	Parents	Money	Decision
Sunny	Yes	Rich	Cinema
Sunny	No	Rich	Tennis
Windy	Yes	Rich	Cinema
Rainy	Yes	Poor	Cinema
Rainy	No	Rich	Stay In
Rainy	Yes	Poor	Cinema
Windy	No	Poor	Cinema
Windy	No	Rich	Shopping
Windy	Yes	Rich	Cinema
Sunny	No	Rich	Tennis

3.A. 1. Explain the concepts of weights, bias, threshold, learning rate, and momentum factor in the training of a neural network. **5**

3.B. (Attempt Any 1 Question) **10**

i. For Bayesian Network Given Below, **10**



Given

P(PowerFailure) = 0.03

P(BatteryLow) = 0.15

P(SystemDown | PowerFailure, BatteryLow) = 0.99

P(SystemDown | PowerFailure, ¬BatteryLow) = 0.95

P(SystemDown | ¬PowerFailure, BatteryLow) = 0.40

P(SystemDown | ¬PowerFailure, ¬BatteryLow) = 0.02

Compute P(BatteryLow|SystemDown)

ii. Classify x= (Color = Yellow, Type = SUV, Origin = Imported) using Naïve Bayes, Given, **10**

Example	Color	Type	Origin	Stolen?
1	Red	Sport	Domestic	Yes
2	Red	Sport	Domestic	No
3	Red	Sport	Domestic	Yes
4	Yellow	Sport	Domestic	No
5	Yellow	Sport	Imported	Yes
6	Yellow	SUV	Imported	No
7	Yellow	SUV	Imported	No
8	Yellow	SUV	Domestic	No
9	Red	SUV	Imported	Yes
10	Red	Sport	Imported	Yes

4.A. Explain how K-Medoids overcomes the limitations of K-Means in handling outliers. Give a suitable example. **5**

4.B. (Attempt Any 1 Question) **10**

i. Given is the distance matrix. Apply the single linkage and average linkage techniques to find clusters using Euclidean distance. Draw dendrograms. Find the number of clusters when threshold = 0.3 and threshold =0.6. **10**

	P1	P2	P3	P4	P5
P1	0.00	1.04	0.59	0.46	0.82
P2	1.04	0.00	0.77	0.62	0.32
P3	0.59	0.77	0.00	0.30	0.45
P4	0.46	0.62	0.30	0.00	0.36
P5	0.82	0.32	0.45	0.36	0.00

ii. Apply K Means algorithm for the data given, up to two iterations and write the clusters. Initially choose the first two points as initial centroids. Take K=2. **10**

Sr. No	X	Y
1	185	72
2	170	56
3	168	60
4	179	68
5	182	72
6	188	77